

MTH401 Grand quiz 2020 (Spring)

MTH401: Grand Quiz		Quiz Start Time: 08:01 AM, 06 July 2020
Question # 1 of 30 (Start time: 08:01:15 AM, 06 July 2020)		Total Marks: 1
The gravitational force exerted by the earth on a body of mass m is called ——— of the body.		
Select the correct option		
☐	Force	
☐	weight	

Question # 2 of 30 (Start time: 08:01:29 AM, 06 July 2020)

Total Marks: 1

The differential equation

$$\frac{du}{dx} + \frac{2}{x}u = 3 - \frac{6}{x}$$

Select the correct option

[Reload Math Equations](#)



$$\frac{1}{x}$$



$$x^2$$



$$\frac{1}{x^2}$$



$$-\frac{1}{x}$$

Maths 1 Quiz

Quiz start time: 08:01 AM, 06 July 2020

Question # 3 of 30 (Start time: 08:01:43 AM, 06 July 2020)

Total Marks: 1

If the auxiliary equation $m^3 + m = 0$ has roots $m = 0, m = \pm i$ then the complementary function is

Select the correct option

[Reload Math Equations](#)



$$y_c = c_1 + c_2 \cos x + c_3 \sin x$$



$$y_c = c_1 \cos x + c_2 \sin x$$

Question # 4 of 30 (Start time: 08:01:48 AM, 06 July 2020)

Total Marks: 1

The differential equation

$$\frac{dy}{dx} - y = y^3$$

Select the correct option

[Reload Math Equations](#)

☐	Bernoulli's
☐	Homogeneous
☐	Cauchy
☐	Bessel

Question # 5 of 30 (Start time: 08:02:26 AM, 06 July 2020)

Total Marks: 1

What is annihilator operator of the function

$$g(x) = 4 \sin x$$

?

Select the correct option

[Reload Math Equations](#)

☐	$(D^2 - 1)$
☐	$(D^2 + 4)$
☐	$(D^2 - 4)$
☐	$(D^2 + 1)$

The differential equation $(\sin 2x - \tan y)dx - x \sec^2 y dy = 0$ is exact because - - - - -.

Select the correct option

[Reload Math Equations](#)

- | | |
|----------------------------------|---|
| ☒ | $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = -\sec^2 y$ |
| ☐ | $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y} = -\sec^2 y$ |
| ☐ | $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = \sec^2 y$ |
| ☐ | $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y} = \sec^2 y$ |

Which of the following is first order linear equation in unknown variable y?

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|---|
| ☐ | $x \frac{dy}{dx} + (\sin x) y = \cos x$ |
| ☐ | $y \frac{dx}{dy} + (\sin y) x = \cos y$ |
| ☐ | $y \frac{dx}{dy} + (\sin y) x = \cos x$ |
| ☐ | $y \frac{dx}{dy} + (\sin x) x = \cos y$ |

Question # 10 of 30 (Start time: 08:03:06 AM, 06 July 2020)

Total Marks: 1

If initial amount of a radioactive isotope is 100g. What will be the amount at the end of 30 days such that $K=0.0437$?

Select the correct option

- | | | |
|-----------------------|---------|--|
| ☐ | 371.415 | |
| ☐ | 380.560 | |
| ☐ | 363.279 | |
| ☐ | 360.351 | |

Question # 11 of 30 (Start time: 08:03:13 AM, 06 July 2020)

Total Marks: 1

For $f(x, y) = x^2 - y^2$, $f(tx, ty) = \dots$

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|---------------|
| ☐ | $f(x, y)$ |
| ☐ | $tf(x, y)$ |
| ☐ | $t^2 f(x, y)$ |
| ☐ | $t^3 f(x, y)$ |

Question # 12 of 30 (Start time: 08:03:19 AM, 06 July 2020)

Total Marks: 1

Which of following would be a constant solution of the separable differential equation: $\frac{dy}{dx} = e^{x+2}$?

Select the correct option

[Reload Math Equations](#)

$y = 0$



$y = 1$



$y = a \in \mathbb{R}$



No constant solution exist

Question # 13 of 30 (Start time: 08:03:24 AM, 06 July 2020)

Total Marks: 1

If we substitute

$$u = \frac{y}{x}$$

in differential equation

Select the correct option

[Reload Math Equations](#)

$ue^u du = dx$



$ue^{(-u)} du = dx$



$ue^u du = dy$



$ue^{(-u)} du = dy$

Question # 14 of 30 (Start time: 08:03:34 AM, 06 July 2020)

Total Marks: 1

Separable form $g(x)dx + f(y)dy = 0$, of the differential equation: $(2x \cos y) dx + x^2 (\sec y - \sin y) dy = 0$ is—.

Select the correct option

[Reload Math Equations](#)

☐ $\left(\frac{2}{x}\right) dx + (\cos^2 y - \tan y) dy = 0$

☐ $\left(\frac{x}{2}\right) dx + (\sec^2 y - \cot y) dy = 0$

☐ $\left(\frac{x}{2}\right) dx + (\sec^2 y - \tan y) dy = 0$

☐ $\left(\frac{2}{x}\right) dx + (\sec^2 y - \tan y) dy = 0$

Question # 15 of 30 (Start time: 08:03:39 AM, 06 July 2020)

Total Marks: 1

A differential equation $M(x,y)dx + N(x,y)dy = 0$ is exact if there exists a multi - variable function $f(x,y)$ such that - - -

Select the correct option

[Reload Math Equations](#)

☐ $df(x,y) = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$

☐ $\int f(x,y) dx = \int \frac{\partial f}{\partial x} dx + \int \frac{\partial f}{\partial y} dy$

☐ $f(x,y) = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$

☐ $f(x,y) = \int \frac{\partial f}{\partial x} dx + \int \frac{\partial f}{\partial y} dy$

$\frac{d^3 y}{dx^3} + y^4 = 0$ is a ——— differential equation of order —.

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|---------------|
| ☐ | linear, 4 |
| ☐ | linear, 3 |
| ☐ | non-linear, 4 |
| ☐ | non-linear, 3 |

Which of the following is an equivalent form of the exact differential equation:
 $ydx + xdy = 0$?

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|---------------------------------|
| ☐ | $d\left(\frac{x}{y}\right) = 0$ |
| ☐ | $d\left(\frac{y}{x}\right) = 0$ |
| ☐ | $d(xy) = 0$ |
| ☐ | $d(x + y) = 0$ |

Question # 18 of 30 (Start time: 08:03:59 AM, 06 July 2020)

Total Marks: 1

$$\frac{dy}{dx} - \frac{1}{xy} = xy^3$$

Identify

Select the correct option

Reload Math Equations

- | | |
|-----------------------|----------------------------------|
| ☐ | $\frac{1}{x}, x \in \mathbb{R}$ |
| ☐ | $y, xy^2 \in \mathbb{R}$ |
| ☐ | $-y, xy^2 \in \mathbb{R}$ |
| ☐ | $-\frac{1}{x}, x \in \mathbb{R}$ |

Question # 19 of 30 (Start time: 08:05:08 AM, 06 July 2020)

Total Marks: 1

The differential equation of orthogonal trajectory to the family of curves

$$x^2 + y^2 = C^2$$

is _____.

Select the correct option

Reload Math Equations

- | | |
|-----------------------|--------------------------------|
| ☐ | $\frac{dy}{dx} = \frac{x}{y}$ |
| ☐ | $\frac{dy}{dx} = \frac{y}{x}$ |
| ☐ | $\frac{dy}{dx} = -\frac{x}{y}$ |
| ☐ | $\frac{dy}{dx} = -\frac{y}{x}$ |

Question # 20 of 30 (Start time: 08:05:17 AM, 06 July 2020)

Total Marks: 1

If the general solution of a separable differential equation is $\sin^{-1} y = \cos^{-1} x + c$, provided that $y(\frac{1}{\sqrt{3}}) = \frac{1}{\sqrt{3}}$, then $c = \dots$.

Select the correct option

[Reload Math Equations](#)

☐	$\frac{\pi}{6}$
☐	$\frac{\pi}{3}$
☐	$-\frac{\pi}{6}$
☐	0

[Click to Save Answer & Move to Next Question](#)

08:05:17 AM, 06 July 2020

08:05:17 AM, 06 July 2020

Question # 21 of 30 (Start time: 08:05:22 AM, 06 July 2020)

Total Marks: 1

The equation of free un-damped motion is _____.

Select the correct option

[Reload Math Equations](#)

☐	$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$
☐	$\frac{d^2x}{dt^2} - \frac{k}{m}x = 0$

Question # 22 of 30 (Start time: 08:05:35 AM, 06 July 2020)

Total Marks: 1

Which of the following is a boundary value problem(BVP)?

Select the correct option

 Reload Math Equations

- | | |
|-----------------------|---|
| ☐ | $\frac{d^2y}{dx^2} + y = 0$ |
| ☐ | $\frac{d^2y}{dx^2} + y = 0, x \in [-2, 2]$ |
| ☐ | $\frac{d^2y}{dx^2} + y = 0, y(1) = -2$ and $y'(1) = 2$ |
| ☐ | $\frac{d^2y}{dx^2} + y = 0, y(0) = 0$ and $y\left(\frac{\pi}{2}\right) = 2$ |

A differential equation $M(x, y)dx + N(x, y)dy = 0$ is exact if and only if ---.

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---|
| ☒ | $\frac{\partial}{\partial x} M(x, y) = \frac{\partial}{\partial y} N(x, y)$ |
| ☐ | $\frac{\partial}{\partial y} M(x, y) = \frac{\partial}{\partial x} N(x, y)$ |
| ☐ | $\frac{d}{dx} M(x, y) = \frac{d}{dy} N(x, y)$ |
| ☐ | $\frac{d}{dy} M(x, y) = \frac{d}{dx} N(x, y)$ |

Question # 24 of 30 (Start time: 08:05:50 AM, 06 July 2020)

Total Marks: 1

For 2nd order linear non-homogeneous differential equation

$$a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

Select the correct option

[Reload Math Equations](#)

☐	$y'(a) = y'_0, \quad y(b) = y_1$
☐	$y'(a) = y'_0, \quad y'(b) = y'_1$
☐	$y'(a) = y'_0, \quad y'(b) = y'_0$
☐	$y'(a) = y'_0, \quad y(b) = y_1$

Wronskian

$$W(1, e^x)$$

= _____

Select the correct option

[Reload Math Equations](#)

☐	0
☐	1
☐	-1
☐	e^x

For the non-exact differential equation $(x^2y + y) dx - xdy = 0$, if $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = -x$, then the integrating factor is ---.

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|--------------------|
| ☐ | $e^{\frac{1}{x}}$ |
| ☐ | $e^{-\frac{1}{x}}$ |
| ☐ | e^x |
| ☐ | e^{-x} |

Question # 27 of 30 (start time: 06:06:23 AM, 06 July 2020)

Total Marks: 1

Which of following would be a general solution of the differential equation: $\frac{dy}{dx} = 4$?

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|--------------|
| ☐ | $y = 4x + a$ |
| ☐ | $y = ax + 4$ |
| ☐ | $y = 4x + 4$ |
| ☐ | $y = ax + a$ |

Question # 28 of 30 (Start time: 08:06:29 AM, 06 July 2020)

Total Marks: 1

Which of the following substitution will transform the differential equation ; $\frac{dy}{dx} = \frac{y}{x} + \sec\left(\frac{y}{x}\right)$, in to separable form?

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|-------------|
| ☐ | $y = v + x$ |
| ☐ | $y = v - x$ |
| ☐ | $y = vx$ |
| ☐ | $x = vy$ |

The differential equation $\frac{dx}{dy} + \frac{1}{y}x = 2 \sin y$ is first order linear in unknown ---

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|---------------------------|
| ☐ | variable x |
| ☐ | variable y |
| ☐ | multi - variables x and y |
| ☐ | $\frac{dy}{dx}$ |

MT11401 Grand Quiz Quiz Start Time: 05:31 AM, 06 July 2020

Question # 7 of 30 (Start time: 08:02:32 AM, 06 July 2020) Total Marks: 1

Separable form $f(y)dy = g(x)dx$, of the differential equation $y - x \frac{dy}{dx} = a \left(y^2 + \frac{dy}{dx} \right)$ is—.

Select the correct option Reload Math Equations

☐	$\frac{1}{y(1+ay)} dy = \frac{1}{x+a} dx$
☐	$\frac{1}{y(1-ay)} dy = \frac{1}{x+a} dx$
☐	$\frac{1}{y(1+ay)} dy = \frac{1}{x-a} dx$
☐	$\frac{1}{y(1-ay)} dy = \frac{1}{x-a} dx$

Made by Ashii

MT11401 Grand Quiz Quiz Start

Question # 1 of 30 (Start time: 04:58:22 AM, 06 July 2020)

If

$$y = c_1 x^{(-2+\sqrt{2})} + c_2 x^{(-2-\sqrt{2})}$$

is the complementary solution of

Select the correct option

☐	$Ax + B$
☐	$Ax^2 + Bx + C$
☐	$Ax^3 + Bx^2 + Cx + D$
☐	$(Ax^2 + Bx)2x^2$

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Windows Taskbar: Type here to search
Taskbar Icons: File Explorer, Edge, Mail, Store, Settings, Task View, Search, Taskbar

MTH401: Grand Quiz Quiz Start

Question # 2 of 30 (Start time: 04:58:21 AM, 06 July 2020)

The periodic time is given by

Select the correct option

☐	$\frac{\omega}{2\pi}$
☐	$\frac{2\pi}{\omega}$
☐	$2\pi + \omega$
☐	$\frac{\pi}{\omega}$

Raza Academy

Windows taskbar: Type here to search, File Explorer, Edge, Chrome, etc.

MTH401: Grand Quiz Quiz Start

Question # 3 of 30 (Start time: 04:59:24 AM, 06 July 2020)

The hook's law states that the force F is proportional to the _____

Select the correct option

☐	Length
☐	Elongation
☐	Weight
☐	None of these

Raza Academy

Windows taskbar: Type here to search, File Explorer, Edge, Chrome, etc.

Question # 4 of 30 (Start time: 04:58:58 AM, 06 July 2020)

If $x^2 y^3 dx + x^3 y^2 dy = 0$ has the equivalent form as $d\left(\frac{1}{3} x^3 y^3\right) = 0$, then its solution is ...

Select the correct option



$x^3 + y^3 = c$



$x^3 - y^3 = c$



$x^3 y^3 = c$



$\frac{x^3}{y^3} = c$

Raza Academy

Question # 5 of 30 (Start time: 05:00:57 AM, 06 July 2020)

Which of the following function would satisfy: $\frac{dy}{dx} = \frac{dy}{dx} = \dots = \frac{dy}{dx}$?

Select the correct option



$y = Ae^x$



$y = Axe^x$



$y = Ae^{x^2}$



$y = Ax^2 e^x$

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Question # 6 of 30 (Start time: 05:02:04 AM, 06 July 2020)

The differential equation

$$\frac{dy}{dx} - y = y^3$$

Select the correct option



Bernoulli's



Homogeneous



Cauchy



Bessel

Raza Academy

Question # 7 of 30 (Start time: 05:02:40 AM, 06 July 2020)

If the auxiliary equation $S(m^2) + m + 0 = 0$ has roots $S m = 0, m = 1$ then the complementary function is

Select the correct option

 $S(y, x) = (c_1, 1) + (c_2, 2)\cos x + (c_3, 3)\sin x$  $S(y, x) = (c_1, 1)\cos x + (c_2, 2)\sin x$

Raza Academy

Question # 8 of 30 (Start time: 05:04:06 AM, 06 July 2020)

if

$$y = 2 + x$$

, then which of the following is true for it?

Select the correct option

- ☐ Its annihilator operator is D
- ☐ Its annihilator operator is D^2
- ☐ Its annihilator operator is D^3
- ☐ Its annihilator operator is $D+1$

Raza Academy

Question # 9 of 30 (Start time: 05:05:24 AM, 06 July 2020)

How are frequency and period related in simple harmonic motion?

Select the correct option

- ☐ They are directly related
- ☐ Their sum is constant
- ☐ They are inversely related
- ☐ None of the above

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Question # 10 of 30 (Start time: 05:08:43 AM, 06 July 2020)

Which of the following is an equivalent form of the exact differential equation:
 $ydx + xdy = 0$?

Select the correct option

☐	$d\left(\frac{x}{y}\right) = 0$
☐	$d\left(\frac{y}{x}\right) = 0$
☐	$d(xy) = 0$
☐	$d(x + y) = 0$

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Question # 11 of 30 (Start time: 05:08:59 AM, 06 July 2020)

Separable form of the differential equation: $\frac{dy}{dx} = y - 1$ is ----, where $v = y - 1$.

Select the correct option

☐	$\frac{dv}{v} = dv$
☐	$dx = vdv$
☐	$\frac{dv}{v} = dx$
☐	$\frac{dv}{v} = \frac{dv}{x}$

Raza Academy

Question # 12 of 30 (Start time: 05:09:09 AM, 06 July 2020)

In exponential model for the population growth

$$P(t) = P_0 e^{kt} \quad \text{If } k > 0, \text{ then } \lim_{t \rightarrow \infty} P(t) =$$

Select the correct option

☐	0
☐	1
☐	∞
☐	$-\infty$

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Question # 13 of 30 (Start time: 05:10:32 AM, 06 July 2020)

If initial amount of a radioactive isotope is 100g. What will be the amount at the end of 30 days such that $K=0.0437$

Select the correct option

☐	371.415
☐	380.560
☒	363.279
☐	360.351

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Question # 14 of 30 (Start time: 05:11:58 AM, 06 July 2020)

Which of the following is a boundary value problem(BVP)?

Select the correct option



$$\frac{d^2y}{dx^2} + y = 0$$



$$\frac{d^2y}{dx^2} + y = 0, x \in [-2, 2]$$



$$\frac{d^2y}{dx^2} + y = 0, y(1) = -2 \text{ and } y'(1) = 2$$



$$\frac{d^2y}{dx^2} + y = 0, y(0) = 0 \text{ and } y\left(\frac{\pi}{2}\right) = 2$$

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Question # 15 of 30 (Start time: 05:12:23 AM, 06 July 2020)

 $\frac{d^2y}{dx^2} + 5\left(\frac{dy}{dx}\right)^3 - 3y = e^{\sin x}$ is an example of ——— differential equation.

Select the correct option



ordinary linear



ordinary non-linear



partial linear



partial non-linear

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Question # 16 of 30 (Start time: 05:13:40 AM, 06 July 2020)

The integrating factor for the first order linear differential equation : $\frac{dy}{dx} + y \cot x = \sin^2 x$ is -- --

Select the correct option



$\sin x$



$\cos x$



$e^{\sin x}$



$e^{\cos x}$

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Question # 17 of 30 (Start time: 05:15:01 AM, 06 July 2020)

What is annihilator operator of the function

$g(x) = 4 \sin x$

?

Select the correct option



$(D^2 - 1)$



$(D^2 + 4)$



$(D^2 - 4)$



$(D^2 + 1)$

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Question # 18 of 30 (Start time: 05:19:39 AM, 06 July 2020)

Which of following are explicit solutions of the differential equation: $\frac{dy}{dx} = -\frac{x}{y}$.

Select the correct option

- ☐ $y = \pm \sqrt{4+x^2}$
- ☒ $y = \pm \sqrt{-4+x^2}$
- ☐ $y = \pm \sqrt{4-x^2}$
- ☐ $y = \pm \sqrt{-4-x^2}$

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MTH401: Grand Quiz

Quiz Start T

Question # 19 of 30 (Start time: 05:19:25 AM, 06 July 2020)

Classify the following differential equation

$$e^x \frac{dy}{dx} + 2y = 3xy$$

Select the correct option

- ☐ Separable and not linear
- ☐ Linear and not separable
- ☐ Both separable and linear
- ☐ Neither separable nor linear

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Question # 20 of 30 (Start time: 05:19:29 AM, 06 July 2020)

$$\text{For } f(x, y) = \frac{2x}{3y} + 7, f(tx, ty) = \dots$$

Select the correct option

☒	$f(x, y)$
☐	$tf(x, y)$
☐	$t^2f(x, y)$
☐	$t^3f(x, y)$

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Question # 21 of 30 (Start time: 05:20:56 AM, 06 July 2020)

The family of parabolas $y^2 = 4ax$ are solutions of the differential equation: $\frac{dy}{dx} = \frac{2a}{y}$ for _____ value(s) of a .

Select the correct option

☐	infinite
☐	finite
☐	unique
☐	no

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Question # 22 of 30 (Start time: 05:22:15 AM, 06 July 2020)

The differential equation of orthogonal trajectory to the family of curves

$$x - 2y = c$$

is

Select the correct option



$$\frac{dy}{dx} = -\frac{1}{2}$$



$$\frac{dy}{dx} = \frac{1}{2}$$



$$\frac{dy}{dx} = 2$$



$$\frac{dy}{dx} = -2$$

Raza Academy

Question # 23 of 30 (Start time: 05:23:30 AM, 06 July 2020)

Which of the following is first order linear equation in unknown variable y ?

Select the correct option

☐

$$x \frac{dy}{dx} + (\sin x)y = \cos x$$

☐

$$y \frac{dx}{dy} + (\sin y)x = \cos y$$

☐

$$y \frac{dx}{dy} + (\sin y)x = \cos x$$

☐

$$y \frac{dx}{dy} + (\sin x)x = \cos y$$

Raza Academy

Question # 24 of 30 (Start time: 05:24:23 AM, 06 July 2020)

Which of following would be a constant solution of the separable differential equation: $\frac{dy}{dx} = e^{x+2}$?

Select the correct option

☐

$$y = 0$$

☐

$$y = 1$$

☐

$$y = a \in \mathbb{R}$$

☐

No constant solution exist

Raza Academy

Question # 25 of 30 (Start time: 05:25:43 AM, 06 July 2020)

The leading coefficient in the differential equation

$$3x \frac{d^2 y}{dx^2} - 7x \frac{dy}{dx} + 5y = 0$$

Select the correct option

☐ 3☐ 7☐ 5☐ -7

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Question # 26 of 30 (Start time: 05:27:24 AM, 06 July 2020)

If the general solution of a separable differential equation is $\sin^{-1} y = \cos^{-1} x + c$, provided that $y\left(\frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}}$, then $c = \dots$ ☐ $\frac{\pi}{4}$ ☐ $-\frac{\pi}{4}$ ☐ $-\frac{\pi}{2}$ ☐ 0

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Question # 27 of 30 (Start time: 05:29:12 AM, 06 July 2020)

Which of following is an implicit solution of the differential equation: $\frac{dy}{dx} = -\frac{x}{y}$.

Select the correct option

☐

$x + y + 4 = 0$

☐

$x^2 + y^2 - 4 = 0$

☐

$x^3 - y^3 + 4 = 0$

☐

$x^2 - y^2 - 4 = 0$

Raza Academy

Question # 28 of 30 (Start time: 05:31:19 AM, 06 July 2020)

Wronskian of 1 and x^2 , $W(1, x^2) = \underline{\hspace{2cm}}$

Select the correct option

☐

0

☐

2x

☐

x

☐

None of these

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Question # 29 of 30 (Start time: 05:31:47 AM, 06 July 2020)

For the non - exact differential equation $M(x, y)dx + N(x, y)dy = 0$, if $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}$ is a function of x , then the integrating factor is

Select the correct option

function of x alsofunction of y multi - variable function of both x and y 

constant

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Question # 30 of 30 (Start time: 05:33:00 AM, 06 July 2020)

If 2,3,5 are real roots of a differential equation, then the general solution is _____.

Select the correct option



$$y_c = c_1 e^{2x} + c_2 e^{3x} + c_3 e^{5x}$$



$$y_c = c_1 e^{2x} + c_2 e^{-3x} + c_3 e^{5x}$$



$$y_c = c_1 e^{2x} + c_2 e^{-3x} + c_3 e^{5x}$$



$$y_c = \sqrt[3]{(c_1 + c_2 + c_3)e^{2x} + e^{2x}}$$

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Made by Ashii

Integrating factor for the first order linear differential equation : $(x-1)^3 \frac{dy}{dx} + 4(x-1)^2 y = \cos x$ is ----

Select the correct option

- | | |
|----------------------------------|---------------------|
| ☐ | e^{x-1} |
| ☒ | $(x-1)^4$ |
| ☐ | $\ln(x-1)$ |
| ☐ | $e^{\frac{1}{x-1}}$ |

The differential equation $\frac{dy}{dx} = \frac{y}{x}$ is ----.

Select the correct option

- | | |
|----------------------------------|-------------|
| ☒ | Exact |
| ☐ | Non exact |
| ☐ | Non-linear |
| ☐ | Inseparable |

The differential equation $(1 + \ln xy)dx + \left(1 + \frac{x}{y}\right)dy = 0$ is exact because ----

Select the correct option

- | | |
|----------------------------------|---|
| ☐ | $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = \frac{1}{x}$ |
| ☐ | $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y} = \frac{1}{y}$ |
| ☒ | $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = \frac{1}{y}$ |
| ☐ | $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y} = \frac{1}{x}$ |

Which of the following is true about

$$f(x, y) = x^3 - y^3 + 37$$

Select the correct option

- ☒ $f(x, y) \neq t^3 f(x, y)$
- ☐ $f(x, y) = t^3 f(x, y)$
- ☐ $f(x, y) = -t^3 f(x, y)$
- ☐ $f(x, y) = t f(x, y)$

Which of the following are explicit solutions of the differential equation $\frac{dy}{dx} = -\frac{x}{y}$?

Select the correct option

- ☐ $y = \pm\sqrt{4+x}$
- ☐ $y = \pm\sqrt{-4+x^2}$
- ☒ $y = \pm\sqrt{4-x^2}$
- ☐ $y = \pm\sqrt{-4-x^2}$

Which of the following substitution will transform the differential equation $\frac{dy}{dx} = \frac{y}{x} + \sec\left(\frac{y}{x}\right)$, in to separable form?

Select the correct option

- ☐ $y = v + x$
- ☐ $y = v - x$
- ☒ $y = vx$
- ☐ $x = vy$

$\frac{d^2y}{dx^2} + 5\left(\frac{dy}{dx}\right)^3 - 3y = e^{mx}$ is an example of ——— differential equation.

Select the correct option

☒ ordinary linear

☐ ordinary non-linear

☐ partial linear

☐ partial non-linear

$\frac{d^2y}{dx^2} + 5\left(\frac{dy}{dx}\right)^3 - 3y = e^{mx}$ is an example of ——— differential equation.

Select the correct option

☒ ordinary linear

☐ ordinary non-linear

☐ partial linear

☐ partial non-linear

A multi-variable function $f(x, y)$ is said to be homogeneous if $f(tx, ty) = t^n f(x, y)$, where $n \in \mathbb{R}$.

Select the correct option

Submit

☐

$\mathbb{N}$ = Set of naturals

☐

$\mathbb{Z}$ = Set of integers

☐

$\mathbb{Q}$ = Set of Rationals

☒

$\mathbb{R}$ = Set of Reals

Separable form of the differential equation $\frac{dy}{dx} = y - 1$ is —, where $v = y - 1$.

Select the correct option

☐

$$\frac{dv}{v} = dx$$

☐

$$dx = vdv$$

☒

$$\frac{dv}{v} = dx$$

☐

$$\frac{dv}{v} = \frac{dv}{v}$$



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For $f(x, y) = x^2y - xy^3, f(tx, ty) = \dots$

Select the correct option

Submit

- ☐ $f(x, y)$
- ☐ $t f(x, y)$
- ☐ $t^2 f(x, y)$
- ☒ $t^3 f(x, y)$

For the non-exact differential equation $(x^2y + y) dx - xdy = 0$, if $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = -x$, then the integrating factor is ---.

Select the correct option

Submit

- ☐ $e^{\frac{1}{x}}$
- ☐ $e^{-\frac{1}{x}}$
- ☐ e^x
- ☒ e^{-x}

Separable form $f(y)dy + g(x)dx = 0$, of the differential equation: $x \sin y dx + (x^2 + 1) \cos y dy = 0$ is—

Select the correct option

☐ $\tan y dy + \frac{x}{x^2+1} dx = 0$

☒ $\cot y dy + \frac{x}{x^2+1} dx = 0$

☐ $\tan y dy + \frac{x}{x^2-1} dx = 0$

☐ $\cot y dy + \frac{x}{x^2-1} dx = 0$

Separable form $f(y)dy + g(x)dx = 0$, of the differential equation: $x \sin y dx + (x^2 + 1) \cos y dy = 0$ is—

Select the correct option

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☒ $\cot y dy + \frac{x}{x^2+1} dx = 0$

☐ $\tan y dy + \frac{x}{x^2-1} dx = 0$

☐ $\cot y dy + \frac{x}{x^2-1} dx = 0$

Which of following is an example of Homogeneous function?

Select the correct option



$$f(x, y) = \sin\left(\frac{1}{x}\right)$$



$$f(x, y) = \sin\left(\frac{x}{y}\right)$$



$$f(x, y) = \sin x$$



$$f(x, y) = \sin xy$$

The general linear ordinary differential equation of order n is $a_n(x)\frac{d^ny}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \dots + a_1(x)\frac{dy}{dx} + a_0(x)y = 0$, where ———.

Select the correct option



$$a_0(x) = 0$$



$$a_0(x) \neq 0$$



$$a_n(x) = 0$$



$$a_n(x) \neq 0$$



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Which of the following is an initial value problem (IVP)?

Select the correct option

- ☐ $\frac{dy}{dx} + y = 0$
- ☐ $\frac{dy}{dx} + y = 0, x \in [-2, 2]$
- ☒ $\frac{dy}{dx} + y = 0, y(1) = -2$ and $y'(1) = -2$
- ☐ $\frac{dy}{dx} + y = 0, y(0) = -2$ and $y'(\frac{1}{2}) = -2$

For $f(x, y) = x^2y - xy^2, f(tx, ty) = \dots$

Select the correct option

- ☐ $f(x, y)$
- ☐ $t f(x, y)$
- ☐ $t^2 f(x, y)$
- ☒ $t^3 f(x, y)$

Which of the following is an initial value problem (IVP)?

Select the correct option

- ☐ $\frac{dy}{dx} + y = 0$
- ☐ $\frac{dy}{dx} + y = 0, x \in [-2, 2]$
- ☒ $\frac{dy}{dx} + y = 0, y(1) = -2$ and $y'(1) = -2$
- ☐ $\frac{dy}{dx} + y = 0, y(0) = -2$ and $y'(\frac{1}{2}) = -2$

On comparing the differential equation $dy = dx$ with standard form of $M(x,y)dx + N(x,y)dy = 0$, we obtain ----

Select the correct option

- ☐ $M(x,y) = N(x,y)$
- ☒ $|M(x,y)| = |N(x,y)|$
- ☐ $\frac{\partial M(x,y)}{\partial x} \neq \frac{\partial N(x,y)}{\partial y}$
- ☐ $\frac{\partial M(x,y)}{\partial y} \neq \frac{\partial N(x,y)}{\partial x}$

The differential equation $(\sin 2x - \tan y)dx - x \sec^2 y dy = 0$ is exact because -----.

Select the correct option

☒	$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = -\sec^2 y$
☐	$\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y} = -\sec^2 y$
☐	$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = \sec^2 y$
☐	$\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y} = \sec^2 y$

Which of the following substitution will transform the differential equation: $\frac{dy}{dx} = \frac{y}{x} + \sec\left(\frac{y}{x}\right)$, in to separable form?

Select the correct option

☐	$y = v + x$
☐	$y = v - x$
☒	$y = vx$
☐	$x = vy$

The general linear ordinary differential equation of order n is: $a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}} + \dots + a_2(x)\frac{d^2 y}{dx^2} + a_1(x)\frac{dy}{dx} + a_0(x)y = 0$, where $\dots$,

Select the correct option

⌂

- ☐ $a_0(x) = 0$
- ☐ $a_0(x) \neq 0$
- ☐ $a_n(x) = 0$
- ☒ $a_n(x) \neq 0$

For the exact differential equation $(y - 2x - 3) dy = 0$, if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = \frac{2}{y}$, then the integrating factor is $\dots$

Select the correct option

⌂

- ☐ $-\frac{1}{y^2}$
- ☐ $\frac{1}{y^2}$
- ☐ $-y^2$
- ☒ y^2

Which of following is an example of Homogeneous function?

$$\frac{1}{x}$$

Select the correct option

☐

$$f(x, y) = \sin\left(\frac{1}{x}\right)$$

☒

$$f(x, y) = \sin\left(\frac{x}{y}\right)$$

☐

$$f(x, y) = \sin x$$

☐

$$f(x, y) = \sin xy$$

☐

$$f(x, y) = \sin xy$$

The degree of the differential equation $\frac{dy}{dx} + 5\left(\frac{dy}{dx}\right)^2 - 3y = e^{mx}$ is—.

Select the correct option

☐

0

☐

1

☐

6

☒

3

Which of the following would be a particular solution of the differential equation: $\frac{dy}{dx} = -4$?

$$-\frac{1}{4}$$

Select the correct option

- | | |
|----------------------------------|--------------|
| ☒ | $y = 4x + a$ |
| ☐ | $y = ax + 4$ |
| ☐ | $y = ax + a$ |
| ☐ | $y = 4x + 4$ |

Integrating factor for the first order linear differential equation: $(x-1)^3 \frac{dy}{dx} + 4(x-1)^2 y = \cos x$ is: ---

Select the correct option

- | | |
|----------------------------------|---------------------|
| ☐ | e^{x-1} |
| ☒ | $(x-1)^4$ |
| ☐ | $\ln(x-1)$ |
| ☐ | $e^{\frac{1}{x-1}}$ |

Separable form $f(y)dy + g(x)dx = 0$, if the differential equation: $x \sin y dx + (x^2 + 1) \cos y dy = 0$ is—

Select the correct option

☐ $\tan y dy + \frac{x}{x^2+1} dx = 0$

☒ $\cot y dy + \frac{x}{x^2+1} dx = 0$

☐ $\tan y dy + \frac{x}{x^2-1} dx = 0$

☐ $\cot y dy + \frac{x}{x^2-1} dx = 0$

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non-linear, 3

Which of the following is a boundary value problem(BVP)?

Select the correct option

☐ $\frac{dy}{dx} + y = 0$

☐ $\frac{dy}{dx} + y = 0, x \in [-2, 2]$

☐ $\frac{dy}{dx} + y = 0, y(1) = -2$ and $y'(1) = 2$

☒ $\frac{dy}{dx} + y = 0, y(0) = 0$ and $y\left(\frac{1}{2}\right) = 2$

Made by Ashii